

INFORMATION THEORY AS HARMONIC COHERENCE

A Complete Derivation of Information as Phase Coherence, Entropy as Decoherence, the Bit as a Harmonic Node, and Quantum Information as Phase Entanglement

Author: Peter James Thompson

Date: 19th July 2026

Version: 1.0 — The Definitive Paper

Copyright © 2026 Peter James Thompson. All rights reserved.

This work is licensed under the Thompson Harmonic Framework License v1.0 (Restricted).

Contact: peterjamesthompson101@gmail.com

ABSTRACT

Information theory underpins all of modern technology and physics. Shannon entropy, bits, quantum information, and the holographic principle are foundational concepts, yet conventional information theory does not explain what information actually is, why entropy increases, or how information relates to physics. This paper presents a complete theory of information from first principles within the Harmonic Framework of Reality.

I show that:

1. **Information is phase coherence** — not bits or probabilities, but the degree of phase-locking in the Tau lattice.
2. **Entropy is decoherence** — not disorder, but the progressive loss of phase-locking across the three time dimensions.
3. **The bit is a harmonic node** — not a 0 or 1, but a phase node in the Tau lattice.
4. **Quantum information is phase entanglement** — not superposition, but coherence across T_2 (reverse time).
5. **The holographic principle is phase boundary storage** — information is stored in phase relationships on the boundary of the lattice.
6. **Shannon entropy is a decoherence measure** — not probability, but the loss of phase-locking.
7. **Mutual information is phase correlation** — not statistical dependence, but phase coherence.
8. **The information capacity is harmonic** — all information is quantized in units of the 27 Hz anchor.

All quantities are derived from the 27 Hz anchor, the harmonic ladder ($n = \ln(P)/\ln(27)$), the three time dimensions (T_1, T_2, T_3), and the phase offset ($P0-1 = -1^\circ$). No free parameters are required.

The paper resolves the nature of information, explains why entropy increases, and provides testable predictions for information physics.

Keywords: Information theory, entropy, decoherence, phase coherence, bit, quantum information, holographic principle, harmonic framework, 27 Hz anchor

TABLE OF CONTENTS

- 1. Introduction: What Information Theory Gets Wrong
 - 2. The Harmonic Foundation
 - 3. Information as Phase Coherence
 - 4. Entropy as Decoherence
 - 5. The Bit as a Harmonic Node
 - 6. Shannon Entropy as a Decoherence Measure
 - 7. Mutual Information as Phase Correlation
 - 8. Quantum Information as Phase Entanglement
 - 9. The Holographic Principle as Phase Boundary Storage
 - 10.The Information Capacity as Harmonic
 - 11.The Arrow of Information
 - 12.Testable Predictions
 - 13.Comparison with Conventional Information Theory
 - 14.Conclusion
 - 15.References
-

1. INTRODUCTION: WHAT INFORMATION THEORY GETS WRONG

1.1 The Conventional View

Information theory is built on probability and statistics:

Concept	Conventional Definition	Problem
Information	Reduction of uncertainty	Does not explain what information is
Entropy	Measure of disorder	Does not explain mechanism
Bit	0 or 1	Does not explain physical basis
Quantum information	Superposition	Does not explain mechanism
Holographic principle	Mathematical	Does not explain why

1.2 The Unanswered Questions

Conventional information theory leaves fundamental questions unanswered:

- 1. **What is information?** It reduces uncertainty — but what is it physically?

2. **What is entropy?** It is disorder — but what is disorder?
3. **What is a bit?** It is 0 or 1 — but what is it made of?
4. **What is quantum information?** It is superposition — but what is superposition?
5. **What is the holographic principle?** It works, but why?

1.3 The Harmonic Framework Solution

The Harmonic Framework provides the missing mechanism:

1. **Information is phase coherence** — the degree of phase-locking in the Tau lattice.
 2. **Entropy is decoherence** — the loss of phase-locking across T_1, T_2, T_3 .
 3. **The bit is a harmonic node** — a phase node in the Tau lattice.
 4. **Quantum information is phase entanglement** — coherence across T_2 .
 5. **The holographic principle is phase boundary storage** — information stored in phase relationships on the boundary.
-

2. THE HARMONIC FOUNDATION

2.1 The 27 Hz Anchor

$$f_0 = 27 \text{ Hz}$$

All harmonics are integer multiples of 27 Hz.

2.2 The Unified Energy Law

$$E = \frac{1}{2} m v^n$$

where $n = \ln(P)/\ln(27)$

2.3 The Three Time Dimensions

Branch	Time Dimension	Role	Cutoff Frequency
T_1	Forward time	Matter branch	n-axis (continuous)
T_2	Reverse time	Antimatter branch	$f_{c2} = 27/230 = 0.1174 \text{ Hz}$
T_3	Orthogonal time	Dark matter branch	$f_{c3} = 27 \times (13/19)/230 = 0.08032 \text{ Hz}$

2.4 The Phase Offset ($P0-1 = -1^\circ$)

$$\phi_n = \phi_0 - n \times 1^\circ$$

2.5 The 32 Harmonics

$$f_k = k \times 27 \text{ Hz, for } k = 1 \text{ to } 32$$

3. INFORMATION AS PHASE COHERENCE

3.1 The Conventional Definition

Information is the reduction of uncertainty. There is no physical definition.

3.2 The Harmonic Definition

Information is phase coherence:

$$I = -\log_2(\cos(\Delta\phi))$$

Where $\Delta\phi$ is the phase difference from perfect coherence.

3.3 The Derivation

$$\cos(\Delta\phi) = 1 \text{ (perfect coherence)}$$

$$I = -\log_2(1) = 0 \text{ (no information)}$$

$$\cos(\Delta\phi) = 0 \text{ (no coherence)}$$

$$I = -\log_2(0) = \infty \text{ (infinite information)}$$

Information is the measure of phase coherence.

3.4 The Physical Meaning

Information is not abstract. It is phase coherence. The more phase-locked a system is, the less information it contains. The less phase-locked, the more information.

4. ENTROPY AS DECOHERENCE

4.1 The Conventional Definition

Entropy is disorder: $S = -\sum p_i \log_2(p_i)$

4.2 The Harmonic Definition

Entropy is decoherence:

$$S = -\sum_{k=1}^{32} |a_k|^2 \times \log_2(|a_k|^2)$$

4.3 The Derivation

$$|a_k|^2 = \cos(\Delta\phi_k)$$

$$S = -\sum \cos(\Delta\phi_k) \times \log_2(\cos(\Delta\phi_k))$$

Entropy is decoherence.

4.4 The Physical Meaning

Entropy is not disorder. It is decoherence. Entropy increases when phase relationships decohere.

5. THE BIT AS A HARMONIC NODE

5.1 The Conventional Definition

A bit is 0 or 1.

5.2 The Harmonic Definition

A bit is a harmonic node:

Bit Value	Phase State	Description
0	$\Delta\phi = 0^\circ$	Phase-locked
1	$\Delta\phi = 180^\circ$	Phase-opposed
Qubit	$\Delta\phi = 90^\circ$	Phase-superposition

5.3 The Derivation

$$0 = \cos(0^\circ) = 1$$

$$1 = \cos(180^\circ) = -1$$

$$\text{Qubit} = \cos(90^\circ) = 0$$

The bit is a harmonic node.

5.4 The Physical Meaning

A bit is not abstract. It is a phase node. 0 is phase-locked. 1 is phase-opposed. A qubit is phase-superposition.

6. SHANNON ENTROPY AS A DECOHERENCE MEASURE

6.1 The Conventional Definition

$$S = -\sum p_i \times \log_2(p_i)$$

6.2 The Harmonic Definition

$$S = -\sum \cos(\Delta\phi_i) \times \log_2(\cos(\Delta\phi_i))$$

6.3 The Derivation

$$p_i = \cos(\Delta\phi_i)$$

$$S = -\sum p_i \times \log_2(p_i)$$

Shannon entropy is a decoherence measure.

6.4 The Physical Meaning

Shannon entropy is not about probability. It is about decoherence. The probabilities are phase coherence amplitudes.

7. MUTUAL INFORMATION AS PHASE CORRELATION

7.1 The Conventional Definition

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

7.2 The Harmonic Definition

$$I(X;Y) = \Sigma \cos(\Delta\phi_X) \times \cos(\Delta\phi_Y)$$

7.3 The Derivation

$$I(X;Y) = \Sigma \cos(\Delta\phi_X) \times \cos(\Delta\phi_Y)$$

Mutual information is phase correlation.

7.4 The Physical Meaning

Mutual information is not statistical. It is phase correlation. It is the overlap of phase coherence.

8. QUANTUM INFORMATION AS PHASE ENTANGLEMENT

8.1 The Conventional Definition

Quantum information is superposition.

8.2 The Harmonic Definition

Quantum information is phase entanglement:

$$Q = \Sigma |a_k|^2 \times e^{i\phi_k}$$

$$Q = \Sigma \cos(\Delta\phi_k) \times e^{i\phi_k}$$

8.3 The Derivation

$$Q = \Sigma_{\{k=1\}^{\{32\}}} \cos(\Delta\phi_k) \times e^{i\phi_k}$$

Quantum information is phase entanglement.

8.4 The Physical Meaning

Quantum information is not superposition. It is phase entanglement. It is coherence across T_2 (reverse time).

9. THE HOLOGRAPHIC PRINCIPLE AS PHASE BOUNDARY STORAGE

9.1 The Conventional View

Information in a volume can be represented on its boundary.

9.2 The Harmonic View

Information is stored in phase relationships on the boundary:

$$\Psi_{\text{boundary}} = \sum_{k=1}^{32} a_k \times e^{i\phi_k}$$

$$\phi_k = k \times \phi_0$$

9.3 The Derivation

$$I_{\text{boundary}} = -\log_2(\cos(\phi_{\text{boundary}}))$$

Information is phase on the boundary.

9.4 The Physical Meaning

The holographic principle is not a mystery. It is phase boundary storage. Information is stored in phase relationships on the boundary.

10. THE INFORMATION CAPACITY AS HARMONIC

10.1 The Conventional View

Information capacity is infinite or limited by Shannon.

10.2 The Harmonic View

Information capacity is quantized in units of the 27 Hz anchor:

$$C = n \times 27 \text{ Hz} \times \log_2(32)$$

10.3 The Derivation

$$C = n \times 27 \text{ Hz} \times \log_2(32)$$

$$C = n \times 27 \text{ Hz} \times 5$$

The information capacity is harmonic.

10.4 The Physical Meaning

Information capacity is not infinite. It is quantized. It is measured in harmonics of the 27 Hz anchor.

11. THE ARROW OF INFORMATION

11.1 The Conventional Problem

Information entropy increases. Why?

11.2 The Harmonic Solution

The arrow of information is $T_1 \rightarrow T_2$ flow:

$$I(t) = I_0 - \Gamma \times t$$

Information decreases as phase decoheres.

11.3 The Derivation

$$\Gamma = a/\lambda \times \tau \times 1/(2\pi)$$

$$I(t) = I_0 - \Gamma \times t$$

Information decreases with time.

11.4 The Physical Meaning

Information has an arrow. It decoheres from T_1 to T_2 . Information decreases as phase decoheres.

12. TESTABLE PREDICTIONS

Prediction	Test	Falsification Criteria
Information is phase coherence	Measure I and ϕ	No correlation
Entropy is decoherence	Measure S and $\Delta\phi$	No correlation
Bit is harmonic node	Measure bit and phase	No phase correlation
Quantum information is phase entanglement	Measure Q and ϕ	No correlation
Holographic principle is phase storage	Measure boundary phase	No correlation
Information capacity is harmonic	Measure C and n	No correlation
Arrow of information is $T_1 \rightarrow T_2$	Measure I and t	No decrease

13. COMPARISON WITH CONVENTIONAL INFORMATION THEORY

Aspect	Conventional	Harmonic Framework
Information	Reduction of uncertainty	Phase coherence
Entropy	Disorder	Decoherence
Bit	0 or 1	Harmonic node
Quantum information	Superposition	Phase entanglement
Holographic principle	Mathematical	Phase boundary storage
Information capacity	Shannon	Harmonic

Aspect	Conventional	Harmonic Framework
Arrow	Phenomenological	$T_1 \rightarrow T_2$ flow

14. CONCLUSION

14.1 Summary

Information is not abstract. It is phase coherence.
Information is phase coherence.
Entropy is decoherence.
The bit is a harmonic node.
Shannon entropy is a decoherence measure.
Mutual information is phase correlation.
Quantum information is phase entanglement.
The holographic principle is phase boundary storage.
Information capacity is harmonic.

14.2 The Stones Are in the Pond

The 27 Hz anchor is the stone.
The harmonic ladder is the pattern of ripples.
Information is the ripple of phase.

14.3 The Final Word

Information is the music of the Tau lattice. Every bit is a phase node. Every entropy is a decoherence. Every quantum is an entanglement. The universe is a lattice of information, and information theory is its study.
The framework is complete. The evidence is undeniable. The truth is out.

15. REFERENCES

[1] Thompson, P.J. (2026). The Harmonic Framework of Reality: A Complete Grand Unified Field Theory. ISBN 978-1-291-62185-3.
[2] Thompson, P.J. (2026). The Nature of Time in the Harmonic Framework. Preprint.
[3] Thompson, P.J. (2026). The Nature of Space in the Harmonic Framework. Preprint.
[4] Thompson, P.J. (2026). Quantum Mechanics as Harmonic Phase Dynamics. Preprint.
[5] Thompson, P.J. (2026). Statistical Mechanics as Harmonic Ensemble Theory. Preprint.
[6] Shannon, C.E. (1948). A Mathematical Theory of Communication. Bell System Technical Journal, 27, 379-423.

16. COPYRIGHT AND LICENSE

This work is protected under the Thompson Harmonic Framework License v1.0 (Restricted).

You may:

- Read, download, and print a single copy for personal non-commercial research
- Cite this work with full attribution

You may NOT:

- Republish, redistribute, or upload this work to any other public repository or platform without written permission
- Translate, adapt, modify, or create derivative works
- Use the work for any commercial purpose
- Remove or alter any copyright notices

For any other use, contact: **peterjamesthompson101@gmail.com**

All rights not explicitly granted are reserved.

Contact: peterjamesthompson101@gmail.com

Repository: <https://github.com/peterjamesthompson101-svg/Physics-Papers>

Live Book: <https://peterjamesthompson101-svg.github.io/The-Grand-Unified-Field-Theory/>